PeerWatch: Multi-Signal Network Anomaly Detection for Local Area Networks

Darius Chitoroaga

MEng Computer Science

Submission date: TODO: Day Month Year

Supervised by: TODO: Supervisor's Name

Disclaimer: This report is submitted as part requirement for the MEng Computer Science at UCL. It is substantially the result of my own work except where explicitly indicated in the text.

The report may be freely copied and distributed provided the source is explicitly acknowledged.

Abstract —

1 INTRODUCTION

The typical home network today bears little resemblance to the isolated subnet of a decade ago. Driven by the proliferation of connected devices, residential and small-office subnets now carry a mixture of enterprise endpoints, smart home appliances, voice assistants, and networked infrastructure that receives infrequent security updates and is rarely under active administrative oversight [6]. The 2016 Mirai botnet demonstrated at scale that consumer-grade connected devices could be silently weaponised for coordinated global attacks without their owners' awareness [1]; the attack surface in the average home network has only expanded since. Yet the monitoring tools available to independent administrators have not kept pace with this expanded threat surface, and this gap motivates the present work.

Local area networks are built on protocols designed for trusted environments. The Address Resolution Protocol (ARP), which maps IP addresses to MAC addresses on a subnet, carries no authentication: any host can broadcast an ARP reply claiming any binding it chooses [25]. An attacker with network access can silently impersonate another device, redirecting traffic through a machine they control; a man-in-the-middle (MitM) attack that enables credential theft, session hijacking, and traffic inspection without alerting the victim or the network administrator [4].

The standard hardware-enforced defences against ARP spoofing: Dynamic ARP Inspection (DAI) and DHCP snooping, require managed layer-2 switches that maintain a trusted binding table [26]. These features are absent from commodity home and small-office routers, the environments where monitoring is often most needed and least configured. Software-based tools such as arpwatch [16] fill part of this gap by alerting on changes to ARP bindings, but they operate on a single signal. An attacker who compromises a device without forging ARP replies, physically replaces a device, or gradually migrates a service to a new host is entirely invisible to ARP-only monitors.

A more robust approach treats device identity as a composite of independent signals: the operating system inferred from TCP/IP stack behaviour, the set of open ports and running services, the time-to-live (TTL) field of outgoing packets, the route traffic takes through the network, and the cryptographic identity anchors that services such as SSH and TLS expose. Significant drift in any of these signals, or the co-occurrence of smaller drifts across several signals in the same scan cycle, indicates that the device behind an IP address may not be the device previously observed there. When the same anomaly pattern appears simultaneously across multiple hosts, the probability of a coordinated attack rather than a routine configuration change rises substantially. The challenge is calibrating this without drowning operators in noise. Routine events, a firmware update that alters a device's OS fingerprint, a DHCP lease renewal that reassigns an IP, or a router reconfiguration that shifts TTL values, produce observations indistinguishable from an attack in isolation. A monitor that fires on every anomaly is quickly ignored, defeating its own purpose, causing alertfatigue for admins, [14]; careful evidence accumulation, signal weighting, and decay are therefore as important as the signals themselves.

Enterprise network detection and response (NDR) platforms such as Darktrace [9] and Cisco Secure Network Analytics [7] provide sophisticated multi-signal monitoring, but require dedicated appliances, managed infrastructure, or cloud connectivity that place them beyond the reach of home and small-office administrators. Lightweight open-source tools such as arpwatch are deployable on existing hardware but, as noted, are restricted to a single signal. Network intrusion detection systems (NIDS) such as Snort [28] and Suricata [22], and the network analysis framework Zeek [24],

offer deep packet inspection and protocol analysis but operate on signature matching rather than per-device identity drift; they are designed for dedicated network taps and do not maintain a persistent fingerprint model per LAN device. The gap between these categories of tool is summarised in Table 1. No open-source, locally-deployable tool currently integrates active device fingerprinting, passive observation, and fleet-level co-occurrence analysis in a single monitor.

Tool	Active scan	Passive capture	Fleet correl.	Local deploy
Darktrace / Cisco SNA	✓	✓	✓	✗
Snort / Suricata	✗	✓	✗	✓
Zeek	✗	✓	✗	✓
arpwatch	✗	✓ (ARP only)	✗	✓
PeerWatch	✓	✓	✓	✓

Table 1: Detection capability comparison across monitoring tools relevant to small-office administrators.

This project aims to close that gap, building a tool designed specifically for the independent administrator who has neither managed switching infrastructure nor an enterprise security budget.

The threat model assumed throughout this work places the attacker within the local network — through a compromised device, a rogue wireless association, or physical access — and grants them the ability to forge ARP replies, substitute physical devices, or redirect services to attacker-controlled hosts. The attacker is not assumed to perfectly replicate every observable property of the target; the detection strategy exploits precisely this constraint. The monitoring host is assumed to hold a network interface capable of promiscuous-mode capture and sufficient privilege to run periodic `nmap` scans, and is itself assumed to be uncompromised.

1.1 Aims and Goals

The *aim* of this project is to design, implement, and evaluate **PeerWatch**: an open-source, autonomous network anomaly detection daemon that fingerprints subnet devices across multiple independent signals, tracks device identity over time, and flags evidence of spoofing or coordinated attack; deployable on hardware a home or small-office administrator already owns, with no cloud dependencies.

The specific *goals* are:

1. **Active device fingerprinting.** Build a periodic `nmap`-based scan loop [17] that maintains a persistent identity store (`PeerStore`) for each discovered device, tracking OS family, open ports, service types, and MAC address. Define a scoring engine that accumulates suspicion for detected drift between successive scans and decays it exponentially, so stale anomalies do not permanently penalise a device.

2. **Passive monitoring layer.** Supplement active scans with live packet capture that tracks TTL consistency, monitors ARP replies for binding conflicts, performs TCP stack fingerprinting [32], and checks route path stability; detection signals that require no active probing and that catch attacks leaving no nmap-visible trace.

3. **Fleet-level correlation.** Implement a correlator that identifies coordinated anomaly patterns across multiple peers within the same scan tick, such as simultaneous ARP poisoning of several hosts or a subnet-wide route shift, and scores them independently of per-device analysis.

4. **LLM-assisted triage.** Integrate a locally-run large language model that produces structured natural-language investigation reports when a device's suspicion score crosses a threshold, reducing the cognitive burden on the operator who must decide whether to act [14]. Recent advances in locally-runnable models and software allowing models to run on commodity hardware via Ollama [21] have made LLM-assisted triage increasingly viable in isolated environments without cloud infrastructure. No monitored traffic leaves the subnet, no API key or subscription is required, and the system remains functional in air-gapped deployments. The LLM explains triggered alerts and does not participate in the detection decision itself. To prevent indirect prompt injection [13], the model is never given access to raw peer data such as packet payloads, hostnames, or service banners; it receives only structured summaries produced by the detection pipeline, containing numeric scores, event type labels, and device identifiers.

5. **Empirical evaluation.** Measure detection performance across a suite of simulated attack scenarios drawn from MITRE ATT&CK and CVE-documented techniques, and characterise the false positive rate under clean traffic.

1.2 Scope

PeerWatch targets subnets of up to approximately 254 hosts (/24 CIDR blocks) on hardware the administrator already owns: a dedicated monitoring device, repurposed laptop, or single-board computer. The following are explicitly outside scope: WAN-level or inter-subnet threats; decryption of encrypted traffic or deep protocol reconstruction; scalability to multi-subnet enterprise deployments; and detection of attacks originating from the monitoring host itself. The system is designed for deployments where the operator is also the administrator, home users and small offices, not for SOC environments where dedicated appliances and analyst teams are available.

1.3 Approach

PeerWatch was developed iteratively across three phases, each adding a new detection dimension while preserving the correctness of the previous ones. A key insight shaping the architecture was that each phase was motivated by a concrete category of attack the previous layer could not detect: ARP-only monitors are blind to device substitution that bypasses ARP entirely, and active scan profiles alone cannot catch adversaries who replicate nmap-visible behaviour while diverging at the packet level. The interaction between signals also proved more informative than any individual signal, simultaneous drift in OS fingerprint, TTL baseline, and port profile within one scan tick is qualitatively different from any single drift in isolation.

Phase 1 established the core detection loop: periodic `nmap` scans of the target subnet, a `PeerStore` keyed by MAC address (or IP for MAC-less devices), and a rule-based scoring engine that compares each scan against the stored baseline. When a device's score crosses a configurable threshold, a `SuspiciousAgent` invokes a locally-run Ollama model for structured JSON triage, falling back to rule-based severity assignment if the model is unavailable.

Phase 2 added a passive capture loop running alongside the active scan. Passive observations: TTL deviation from an established baseline, ARP reply conflicts, TCP fingerprint inconsistencies, IP ID sequence jumps, and traceroute path changes, accumulate in per-device observation models and contribute to the suspicion score independently of the active scan. This layer catches attacks that exactly mimic a target's nmap-visible profile but cannot replicate every low-level TCP stack behaviour.

Phase 3 introduced the `FleetCorrelator`, which runs each tick after ingestion and before investigation. It detects patterns where multiple peers exhibit the same anomaly simultaneously, applies a configurable suspicion boost to each matching device, and injects fleet-level context into the

LLM prompt, so the agent can reason about coordinated attacks rather than isolated per-device events. Phase 3 also added SSH host-key and TLS certificate fingerprint tracking as high-value cryptographic identity anchors.

Throughout, all scoring weights, thresholds, and daemon settings are exposed through a single Pydantic [8] configuration model, keeping the pipeline transparent and tunable without code changes. The system is designed for subnets of a few hundred hosts on hardware a home or small-office administrator already owns; larger deployments would require scan parallelism.

1.4 Report Structure

Chapter 2 provides the context for this work: background on LAN-layer attack techniques, a survey of related detection tools, and an introduction to the key libraries and components on which PeerWatch is built. Chapter 3 captures the requirements and analyses them into an initial design. Chapter 4 describes the design and implementation of PeerWatch in detail, following the three-phase structure above. Chapter 5 presents the testing strategy and empirical evaluation across the simulated attack suite and a clean-traffic false-positive measurement. Chapter 6 concludes with a critical assessment of the results against the goals set out above, and directions for future work.

2 BACKGROUND AND CONTEXT

2.1 LAN-Layer Attack Techniques

Understanding the threat landscape PeerWatch targets requires examining how attackers exploit the trust assumptions embedded in local network protocols. The attacks described below are real-world examples: each maps to documented CVEs or MITRE ATT&CK techniques [20] and corresponds directly to detection scenarios evaluated in Chapter 5.

2.1.1 ARP Cache Poisoning

The Address Resolution Protocol's trust model was introduced in Chapter 1; this section examines the attack mechanics in sufficient detail to motivate the detection signals that follow.

An ARP poisoning attack proceeds in two steps. The attacker broadcasts gratuitous ARP replies (unsolicited announcements claiming a particular IP-to-MAC binding) targeting both the victim host and the default gateway. Because standard operating system ARP implementations update their cache on receipt of any ARP reply regardless of whether a request was issued [25], both targets update their routing tables to direct traffic through the attacker's machine. The attacker, forwarding traffic between victim and gateway while inspecting it, achieves a man-in-the-middle position transparent to both parties.

From this position an attacker can intercept cleartext credentials, inject content into unencrypted sessions, or strip TLS by presenting a forged certificate to the victim while maintaining a legitimate upstream connection. CVE-2020-25705 (SAD DNS [18]) demonstrates that IP address ownership cannot be treated as self-authenticating even without ARP manipulation; ARP poisoning is the local-network equivalent, documented as MITRE ATT&CK T1557.002 [20].

The MAC address change ARP poisoning produces is detectable by tools such as arpwatch. It is only one observable, however: if the attacker substitutes a device with an identical MAC, or if the attack proceeds through a compromised device that retains its original MAC; the ARP signal is entirely clean, and a richer set of observations is required.

2.1.2 Device Substitution Without ARP Forgery

A more operationally sophisticated attack replaces or impersonates a device without disturbing ARP bindings. The attacker clones the target device's MAC address, trivially accomplished on modern

operating systems, and begins responding to network traffic from attacker-controlled hardware. No ARP anomaly is produced; arpwatch observes nothing.

What does change is the device's observable identity at the network stack level. Different hardware exposes a different TCP/IP stack fingerprint: initial TTL values, IP identification field patterns, TCP window size, window scaling options, and initial sequence number generation are determined by the operating system and kernel version, not the MAC address [17]. An attacker running Windows on hardware that previously hosted a Linux IoT device will produce OS fingerprinting results inconsistent with the stored baseline, a port profile that may differ, and a TTL baseline that diverges from the established per-device average.

MAC spoofing without OS replication is documented as T1564.006 [20]. The limitations of MAC-only monitoring were demonstrated in CVE-2004-0699 [20], where device identity assumptions were exploited precisely because address-layer credentials were not backed by any deeper identity verification.

2.1.3 Service-Level Impersonation

Where device substitution attacks target the network identity of a host, service-level attacks target specific services running on it.

Service backdoors and protocol takeover. An attacker who has compromised a device may install additional services, replace existing ones with attacker-controlled versions, or expose services on non-standard ports. These produce observable port profile drift: new ports appear in subsequent scans, or services on known ports change type. T1543 (Create or Modify System Process) [20] and US-CERT advisory TA17-181A [30] document service persistence techniques used in documented intrusions. More recent examples include CVE-2024-6387 (regreSSHion [27]), a remote code execution vulnerability in OpenSSH that could enable replacement of a running SSH daemon, and CVE-2024-4577, where protocol-level service replacement produced detectable port/protocol mismatches.

Cryptographic identity anchors. SSH servers present a host key fingerprint to clients on each connection; TLS servers present a certificate fingerprint. These are high-entropy, asymmetrically generated values that cannot be forged without the corresponding private key. A device substitution that replaces an SSH or HTTPS server necessarily exposes a different fingerprint to any observer who has recorded the original. This makes host key and certificate tracking qualitatively stronger evidence than heuristic signals: a determined attacker can tune kernel parameters to approximate an OS fingerprint, but presenting a valid SSH host key for a device they do not control is computationally infeasible.

2.1.4 Coordinated Fleet Attacks

The attack patterns above are considered in isolation; in practice, targeted intrusions frequently affect multiple hosts in the same network concurrently. The distinction matters for detection: a single device showing OS fingerprint drift is consistent with a firmware update; four devices showing simultaneous OS drift is not.

Fleet-level patterns observable in real intrusions include gateway substitution campaigns, where the attacker poisons ARP caches across multiple hosts simultaneously to redirect all subnet traffic, subnet-wide route shifts produced by compromising the default gateway, and identity sweeps where multiple devices are replaced in the same window to reduce single-device anomaly visibility. The 2016 Mirai botnet [1] is a documented precedent for coordinated compromise of multiple LAN devices; post-compromise behaviour included systematic service port exposure consistent with the service sweep pattern.

A critical property of coordinated attacks is that per-device analysis may silently miss them: if each individual device scores below the investigation threshold, no alert fires. Fleet-level correlation (detecting that several peers exhibit the same anomaly class within the same observation window) is the mechanism that closes this gap, and is described in detail in Section 4.x.

2.1.5 Evasion and the Attacker's Observable Constraint

The detection strategy throughout PeerWatch rests on the assumption stated in Chapter 1: an attacker cannot perfectly replicate every observable property of the device they are impersonating. This section examines where that assumption is strong and where it is not.

A determined attacker can replicate several signals. Port profile can be matched by opening the same ports; service type labels can be mimicked by running compatible stacks; OS fingerprint can be approximated through kernel parameter tuning — adjusting TCP initial TTL, window size, and timestamp options. Scenario I in the evaluation (T1036.004 masquerading, CVE-2024-3094) demonstrates this boundary: an attacker who carefully mirrors the target's service profile produces a suspicion score of 0.0, a correctly identified limitation of the active-scanning layer.

What is structurally hard to replicate: TTL baselines accumulated over many scan cycles, where hardware-level routing differences compound; TCP/IP stack microvariances below the granularity of kernel parameter tuning; IP identification field counter patterns, which reflect packet generation rate and hardware; and cryptographic identity anchors, where forging a host key or certificate requires possession of the private key.

The practical consequence is that detection reliability increases with the number of independent signals covered. An attacker who evades the nmap OS fingerprint signal may still expose TTL drift; one who suppresses TTL drift may expose a host key change. No single evasion technique neutralises all signals simultaneously without perfectly replicating the target device; which is, by definition, device substitution rather than impersonation.

2.2 Device Fingerprinting

Device fingerprinting derives a stable identity for a networked host from observable protocol behaviour, without relying on self-reported credentials. This section surveys the techniques PeerWatch draws on, their mechanisms, and their limitations. Chapter 4 describes how each is implemented within the pipeline.

2.2.1 Active Fingerprinting

Active fingerprinting sends crafted probe packets to a target and analyses the responses. Nmap's OS detection engine [17] sends a battery of TCP, UDP, and ICMP probes designed to elicit responses that vary by operating system implementation: a SYN to an open port, a SYN to a closed port, a FIN to an open port, and several ICMP variants, each exposing differences in TCP initial sequence number generation, window size and scaling, timestamp option presence, don't-fragment bit handling, and IP identification field behaviour. The response profile is matched against a database of approximately 5,000 OS fingerprints; a confident match returns the OS family and version, a low-confidence match returns the closest candidates with a probability score.

Beyond OS family, the set of open ports constitutes a stable per-device identity vector. Jaccard similarity: the size of the intersection divided by the size of the union of two port sets, quantifies drift between successive scans. A score below a configurable threshold (0.6 by default) flags meaningful port profile change. Service version detection supplements this: nmap's banner-grabbing identifies the service type on each open port, catching cases where the port number is unchanged but the responding application is not.

Active fingerprinting has well-known limitations. Root privilege is required to send raw packets; probe traffic is visible to network monitors; and OS detection accuracy degrades in virtualised or containerised environments where the guest stack is partially masked by the hypervisor. Most importantly for this work, OS fingerprint can be partially spoofed through kernel parameter tuning, adjusting TCP initial TTL, window size, and timestamp options, as established in Section 2.1.5. These limitations motivate the passive layer described next.

2.2.2 Passive Fingerprinting

Passive fingerprinting derives OS and behavioural signals from traffic already on the wire, without sending probes. The p0f tool [32] pioneered this approach: SYN packets from TCP initiators expose the sender's TTL, TCP window size, window scaling factor, maximum segment size, timestamp echo, and IP option fields: a profile that differs reliably across OS families without any active interaction.

The most operationally significant passive signal for per-device identity tracking is the TTL baseline. Each OS family initialises outgoing packets with a characteristic TTL (Linux: 64, Windows: 128, Cisco IOS: 255); after traversing a fixed number of hops to the monitoring host, the observed value is stable for a given device. A single TTL observation is unreliable, routing changes and transient variation exist, but a per-device baseline accumulated over many observations is robust. Deviation of more than 15 from the established baseline is a reliable indicator of either a routing path change or a device substitution.

The IP identification field provides a complementary signal. Older Linux kernels and many embedded devices generate sequential IP ID values, producing a monotonically increasing counter visible in captured packets. Modern Linux and Windows randomise the field. A transition from sequential to random generation, or a counter jump inconsistent with the device's observed packet rate, indicates a stack change.

Passive observation has a structural advantage: it cannot easily be suppressed by an attacker without controlling all outgoing traffic from the compromised device. Its limitation is the inverse: devices generating no observable traffic are invisible, and promiscuous-mode capture requires the monitoring interface to see the relevant frames.

2.2.3 Cryptographic Identity Anchors

Cryptographic identity anchors are high-entropy, asymmetrically generated values that services expose on connection and that cannot be forged without the corresponding private key.

SSH servers generate a host key pair at installation time and present the public key fingerprint to every connecting client. TLS servers present a certificate fingerprint (the hash of the DER-encoded certificate) on each connection. Both values are stable across reboots and software updates; they change only on explicit reconfiguration or reinstallation. A device substitution that replaces an SSH or HTTPS server necessarily exposes a different fingerprint to any observer who recorded the original, making key or certificate change near-certain evidence of server substitution rather than routine maintenance.

The qualitative difference from heuristic signals is significant. OS fingerprint, TTL, and port profile can all be partially spoofed through configuration; presenting a valid SSH host key for a device the attacker does not control requires breaking RSA or ECDSA — computationally infeasible. A single host key change event therefore carries far more evidential weight than a single OS fingerprint drift.

Internet-scale infrastructure monitoring tools such as Censys [10] apply the same principle across the public internet, indexing SSH and TLS fingerprints to track infrastructure changes and detect certificate substitution. PeerWatch applies the technique at LAN granularity, where the baseline is per-device and changes are flagged in real time against a persistent identity store.

2.2.4 Route Stability and MAC OUI Correlation

Route stability. Traceroute maps the sequence of IP hops between the monitoring host and a target device. For a LAN device with a fixed default gateway, this path is constant; a new intermediate hop or changed hop count indicates a routing table manipulation or gateway substitution, consistent with the fleet patterns described in Section 2.1.4. Team Cymru's IP-to-ASN mapping attributes each hop to an autonomous system; a new ASN in the path indicates traffic leaving via an unexpected upstream provider.

MAC OUI correlation. The first three octets of a MAC address identify the hardware vendor registered with the IEEE (the Organizationally Unique Identifier). Cross-checking OUI vendor against nmap-detected OS family catches a class of substitution neither signal alone would flag: an Apple OUI combined with a Linux OS detection indicates the claimed hardware identity is inconsistent with the network stack behaviour.

This signal carries elevated false positive risk. MAC address randomisation, enabled by default in iOS 14+ and Android 10+, assigns locally-administered addresses per network, making OUI attribution unreliable for mobile devices. Virtual machine adapters expose hypervisor vendor OUIs legitimately inconsistent with the guest OS. OUI correlation is therefore treated as a supporting signal rather than a primary detection trigger.

2.3 Related Work and Tool Survey

Section 2.1 established the attack landscape and Section 2.2 the detection signals. This section surveys existing tools and research, explains why each fails to address the gap PeerWatch targets, and positions PeerWatch within the literature. The capability comparison in Table 1 provides a structured summary; this section provides the qualitative analysis behind it.

2.3.1 Enterprise Network Detection and Response

Enterprise NDR platforms represent the most capable end of the monitoring spectrum. Darktrace [9] builds unsupervised machine learning baselines per device, detecting anomalous behaviour relative to each device's individual norm, a conceptual approach similar in spirit to PeerWatch's per-device identity tracking. Cisco Secure Network Analytics [7] analyses NetFlow traffic metadata to detect volumetric and behavioural anomalies across managed infrastructure. Both platforms offer genuine multi-signal detection with documented effectiveness against advanced persistent threats in enterprise environments.

The gap they do not address is deployment model, not technical capability. Darktrace requires cloud connectivity for model updates and investigation tooling, and its detection logic is proprietary and not user-inspectable. Cisco Secure Network Analytics requires Cisco managed switching infrastructure or a cloud licence. Neither platform runs on commodity hardware; neither is available without enterprise licensing. The threat model they address (enterprise perimeter, managed infrastructure, dedicated security operations staff) differs substantially from the one PeerWatch targets.

2.3.2 Network Intrusion Detection Systems (NIDS)

Snort [28] and Suricata [22] are signature-based NIDS that evaluate each packet or flow against a rule set. They are effective at detecting known attack patterns against published rule sets and are widely deployed in enterprise and research networks. Their fundamental limitation here is that they maintain no per-device identity model: each packet is evaluated in isolation against static rules. Snort can detect ARP spoofing if an ARP rule is deployed, but has no concept of "this device previously ran Windows and now runs Linux", the drift-based detection that characterises PeerWatch's approach.

Zeek [24] is a more flexible protocol analysis framework that supports custom scripting for detection logic. A sufficiently motivated operator could script per-device fingerprint tracking in Zeek, but this is not available out of the box and requires deep domain expertise; Zeek's deployment model also assumes a dedicated high-throughput network tap rather than the edge hardware SOHO (Small Office/Home Office) administrators own. No maintained open-source Zeek module provides the LAN-device identity-drift detection PeerWatch implements.

All three are fundamentally **reactive**: they detect known or statistically anomalous patterns against a current observation. None accumulates a persistent per-device fingerprint baseline that enables temporal identity drift to be measured across scan cycles.

2.3.3 Single-Signal Lightweight Tools

Arpwatch [16] monitors ARP bindings and alerts on new or changed IP-to-MAC mappings. It is lightweight, reliable, and well-suited to the SOHO deployment model. Its limitation is precisely what Section 2.1.2 establishes: it is blind to device substitution that does not disturb ARP bindings, to service-level impersonation, and to coordinated attack patterns that unfold above the ARP layer. XarpS and similar tools share this scope.

Hardware-enforced equivalents, Dynamic ARP Inspection and DHCP snooping [26], are more robust for the ARP signal specifically but require managed layer-2 switching infrastructure absent from SOHO deployments.

Every existing lightweight tool covers a single signal. The design space for a multi-signal, locally-deployable, open-source LAN identity monitor combining active scanning, passive capture, and fleet correlation is effectively vacant.

2.3.4 Academic Work on Device Fingerprinting

A substantial body of academic work addresses device **type** identification from passive network observations. IoT Sentinel [19] identifies IoT device types from DHCP fingerprints and network flow metadata, enabling automatic policy enforcement on home routers. Related work uses DNS query patterns, flow statistics, and protocol behaviour to distinguish device categories for network management and access control.

This work is related but addresses a different question. Classification asks: **what type of device is this?** Identity drift detection asks: **is this the same device as before?** A classifier is not sensitive to identity drift: a compromised device replicating its device type's traffic profile would be correctly classified while going undetected, and a substitution that preserves device type would be invisible. The two approaches are complementary rather than competitive.

Statistical anomaly detection: flow entropy analysis, traffic volume baselines, protocol distribution modelling, underpins commercial NDR systems and has a deep research literature. These operate at aggregate traffic level, making them sensitive to volumetric shifts but not to low-and-slow device substitution that preserves traffic volume while changing device identity.

Direct prior work on per-device multi-signal identity drift detection in open-source LAN monitoring is, to the author's knowledge, sparse. This gap is the primary motivation for PeerWatch.

2.3.5 LLM-Assisted Security Triage

The application of large language models to security operations has grown rapidly since the emergence of capable instruction-following models. Commercial products including Microsoft Copilot for Security and Google Chronicle AI incorporate LLMs for alert summarisation, threat intelligence synthesis, and first-pass triage of SIEM events, targeting the alert fatigue problem [14] that afflicts security operations teams at scale.

Deploying LLMs in security contexts introduces a specific risk: indirect prompt injection [13]. Where an LLM processes data originating from an adversary controlled source, packet payloads, device-reported hostnames, service banners, an attacker can embed instructions that manipulate the model's output. Greshake et al. demonstrate this against several deployed LLM-integrated applications, including cases where injected instructions cause the model to suppress or distort its report to the user. In a network monitoring context the threat is direct: a compromised device could craft hostnames or service banners that cause the triage LLM to downplay the anomaly it is investigating; more importantly, since PeerWatch runs with elevated privileges prompt injection could even compromise a machine.

PeerWatch addresses this with an architectural boundary described in Section 1.2: the LLM receives only structured summaries from the detection pipeline: numeric scores, event type labels, device identifiers, never raw packet payloads or device-supplied strings. The model operates in an explanation role, producing a structured investigation report; it plays no part in the detection or scoring decision. This design adds interpretability without introducing a new attack surface.

2.4 Tools and Libraries

This section introduces non-obvious tool choices and the rationale behind each. Per the scope stated in Section 1.3, PeerWatch targets hardware a SOHO administrator already owns; tool selection reflects that constraint throughout.

2.4.1 Active Scanning: nmap

Nmap [17] is the active scanning engine for all device discovery and fingerprinting. The choice warrants justification against two faster alternatives.

Masscan [12] is capable of scanning the entire IPv4 address space in under six minutes but performs only port discovery: it carries no OS fingerprint database, no service version detection, and produces no OS confidence scores. On a /24 subnet scanned every five minutes, scan speed is irrelevant; fingerprint depth is not. ZMap [11] is an internet-scale single-packet scanner with the same limitation. Both tools are optimised for breadth; nmap is optimised for depth, which is what per-device identity tracking requires.

Nmap's value in this context is threefold. Its OS detection engine matches response profiles against a database of approximately 5,000 fingerprints, returning both a matched OS family and a confidence score that PeerWatch records alongside the result. Service version detection via banner-grabbing identifies the application behind each open port, enabling service-type drift to be detected independently of port number. Structured XML output carries host metadata, port state, service attributes, and OS match candidates in a single machine-readable document, which PeerWatch's parser (`src/peerwatch/parser.py`) consumes directly rather than relying on a thin wrapper library: full control over confidence fields and timing data is preserved.

2.4.2 Passive Capture: Scapy

Passive fingerprinting requires live packet capture with per-field access to ARP, IP, and TCP headers. Four Python libraries were evaluated.

Raw libpcap via ctypes. Libpcap [15] is the standard C library for low-level packet capture on POSIX systems; it provides a file descriptor over a network interface in promiscuous mode and a BPF (Berkeley Packet Filter) compilation interface for pushing kernel-level packet filters to the capture path. BPF filters are boolean expressions over packet fields (i.e. `arp or (tcp and not port 22)`) that are compiled to a bytecode program executed in the kernel before any packet reaches userspace, reducing the volume of data Python must handle. Accessing libpcap from Python requires ctypes, the standard library module for calling C shared libraries via foreign function interface without writing a C extension. The combination provides maximum throughput but demands significant

boilerplate: manual struct layout, pointer arithmetic, and per-protocol field extraction. Appropriate for a production high-throughput capture engine, but not for a research implementation where correctness and readability matter more than throughput on a LAN at the scale we are targeting.

Pyshark. Pyshark [23] wraps tshark [31] — the command-line version of Wireshark, a widely used protocol analyser — as a subprocess, feeding it packets and deserialising its JSON output per packet. The subprocess boundary introduces external process overhead and serialisation cost on every packet; suitable for offline PCAP (packet capture file) analysis, not for real-time per-packet processing in a daemon loop.

Dpkt. Dpkt [29] is efficient for offline PCAP file parsing with a clean Python API, but its live capture support is limited, and it is less expressive than scapy for protocol composition or custom layer access patterns.

Scapy. Scapy [3] is a Python-native packet library where each protocol layer is a first-class Python object, and field access is direct (`pkt[ARP].psrc`, `pkt[IP].ttl`, `pkt[TCP].window`) exactly what passive fingerprinting requires. It supports BPF filter compilation at the capture level, limiting what reaches Python-level handling. Scapy is established in security research with broad protocol coverage and active maintenance. The expressiveness advantage over alternatives outweighs the modest throughput overhead at LAN traffic volumes.

2.4.3 LLM Integration: Ollama, Phi-4, and LangChain

Three related choices form the LLM integration layer.

Ollama [21] is a local inference server that runs quantised open-weights models on commodity hardware via a REST API compatible with the OpenAI client interface. It was selected over direct cloud API providers (OpenAI, Anthropic) because network monitoring context, even in summarised form, must not leave the monitored subnet. An air-gapped deployment must remain functional with no outbound connections; Ollama satisfies this constraint without code changes.

Phi-4 (Microsoft, 14B parameters) is the default model. At 4-bit quantisation it requires approximately 8 GB of RAM, within reach of a dedicated monitoring device the administrator already owns. It was selected over Llama 3 8B (lower resource requirement but weaker structured JSON output reliability observed during development) and Mistral 7B (similar tradeoff) on the basis of structured output quality and instruction-following consistency, both critical for producing well-formed `InvestigationReport` JSON on every invocation. The MIT licence is compatible with the project's open-source goals. The model is configurable; any Ollama-supported model can be substituted via `config.json`.

LangChain [5] provides the `format="json"` grammar-constrained generation interface, ensuring the model produces valid JSON regardless of content, enabling direct deserialisation into a typed Pydantic schema. The alternative of raw HTTP to Ollama's `/api/generate` endpoint was rejected because LangChain's structured output handling, retry logic, and streaming/non-streaming abstraction are non-trivial to reimplement correctly; the investigation report schema is validated against a Pydantic model on deserialisation, and the integration between LangChain's structured output and Pydantic's validation is the key value. If Ollama is unavailable, `SuspiciousAgent` falls back to rule-based severity assignment without invoking LangChain, the dependency is isolated to the LLM path.

2.4.4 Configuration and Validation: Pydantic

All scoring weights, thresholds, and daemon settings are declared in a single Pydantic `PeerWatchConfig` model [8]. Pydantic was chosen over plain dicts or dataclasses for three reasons: automatic type coercion (a JSON string `"3.0"` is coerced to `float` transparently), field-level validation with informative error messages on misconfiguration (a user who sets `suspicion_threshold:`

"high" receives an immediate type error rather than a runtime crash deep in the scoring engine), and self-documenting field types and defaults.

The config.example.json pattern documents all fields with their defaults; users copy it to config.json and override only what they need. model_validate handles partial configs, unset fields use the declared defaults. Every component that consumes a threshold receives it from the same config object, eliminating scattered magic numbers and the risk of drift between independently maintained constants.

2.5 Evidence Accumulation and Anomaly Scoring

The signals described in Section 2.2 are individually noisy: a single TTL deviation is consistent with a transient routing change; a single OS fingerprint shift is consistent with a firmware update. Turning noisy per-signal observations into reliable detection requires a principled accumulation model. This section establishes the theoretical basis; Chapter 4 describes the implementation.

2.5.1 Weighted Scoring

Not all signals carry equal evidential weight. A changed SSH host key is near-certain evidence of server substitution; a Jaccard port profile drift below 0.6 is consistent with routine service changes. Assigning uniform weight to all signals would cause low-evidence events to dominate the score when they fire frequently, drowning higher-confidence signals in noise.

Axelsson [2] demonstrates that in intrusion detection systems the false positive rate of individual signals has an outsized effect on overall system utility: even at high true positive rates, a signal with a 1% false positive rate will produce more false alerts than true ones on a typical network where attacks are rare. The practical implication is that signal weights should reflect each signal's discriminative power: its ability to distinguish attack from benign change in the target deployment. PeerWatch assigns weights by evidential strength: cryptographic anchor changes (+3.0 for port/protocol mismatch, +3.0 for SSH host key change) carry far more weight than heuristic signals (+0.5 for Jaccard drift, +0.5 for MAC conflict) that fire routinely under legitimate operation.

2.5.2 Signal Compositing

Independent signals with low individual discriminative power can be combined to achieve high discriminative power. If two signals each have a 10% false positive rate and fire independently, the joint probability of both firing simultaneously on a benign event is approximately 1%, a tenfold reduction in false positive rate with no loss of true positive rate for attacks that trigger both.

This compositing effect is the formal justification for PeerWatch's threshold design: an OS fingerprint drift alone may not cross the investigation threshold; OS drift combined with TTL deviation and port profile change in the same tick reaches the threshold while remaining unlikely under benign conditions. Treating signals as independent is a conservative approximation, in reality, correlated signals (e.g., OS change and TTL change both caused by device substitution) provide less additional evidence than independent signals would, so the approach underestimates joint evidence rather than overstating it.

2.5.3 Exponential Decay and the Cold-Start Period

A device that triggers anomaly events during a legitimate firmware update should not remain permanently penalised once the update is complete and the new baseline is stable. Exponential decay, where the suspicion score halves every T days, implements a forgetting mechanism: unreinforced anomalies fade over time without a hard reset. Event-window approaches that reset scores after a fixed period produce cliff effects (a device near the window boundary retains full score; one just past it retains none); continuous decay avoids this discontinuity. The half-life parameter (suspicion_half_life_days, default 3.5 days) is tunable to the deployment context.

The cold-start problem, first observed in recommender systems and fraud detection, applies directly to device identity tracking. A device seen for the first time has no baseline; scoring drift against a single observation produces high false positive rates on legitimate newly-discovered devices. PeerWatch withholds scoring for the first `baseline_min_scans` observations (default 5), recording events without contributing to the suspicion score until a stable baseline is established.

2.5.4 Threshold Calibration

The investigation threshold determines the operating point on the precision-recall curve. Lowering the threshold increases true positive rate at the cost of false positive rate; raising it reduces false alerts at the risk of missed detections. Axelsson [2] shows that for realistic base rates of attacks on a typical network, even modest false positive rates render a detection system impractical; an alert that fires ten times per day on a clean network will be ignored within a week, eliminating any security value.

Threshold calibration is therefore not a parameter to set once and forget but a deployment-specific tradeoff between operator tolerance for false alarms and the acceptable miss rate for real attacks. Chapter 5 characterises PeerWatch's operating point empirically under both simulated attack traffic and clean-traffic conditions, providing the data an operator needs to adjust the threshold for their specific environment.

This chapter has established the foundations on which PeerWatch is built. Section 2.1 defined the threat landscape — the attack techniques that existing single-signal tools cannot detect and that motivate a multi-signal approach. Section 2.2 surveyed the fingerprinting signals available to a locally-deployed monitor, from active nmap probing to passive TCP stack observation and cryptographic anchors. Section 2.3 positioned PeerWatch against existing tools and research, identifying the vacant design space it occupies. Sections 2.4 and 2.5 introduced the tools the implementation relies on and the theoretical basis for combining their outputs into a coherent scoring model. Chapter 3 translates this context into a structured requirements specification and analyses it into an initial design.

3 REQUIREMENTS AND ANALYSIS

3.1 Problem Statement

The introduction established that existing tools fail to detect device substitution and service-level impersonation attacks on home and small-office networks. This section states the precise technical problem an implementation must solve: the specific uncertainties, constraints, and failure modes from which the requirements in Section 3.2 are derived.

The system must detect when a device observed at a given MAC or IP address in a previous scan is no longer the same physical or logical device. It must do this without external ground truth: at runtime there is no oracle to confirm whether a fingerprint change represents a firmware update or an attack. Detection is therefore probabilistic, evidence accumulates across scan cycles until a configurable confidence threshold is crossed, at which point investigation is triggered. The goal is not to prove an attack has occurred, but to surface sufficient evidence that an operator or autonomous rule can act on it with an acceptable false positive rate. This distinguishes the problem from signature-based intrusion detection, which matches observations against known attack patterns; PeerWatch detects drift against a self-built per-device baseline regardless of whether the underlying technique appears in any signature database.

Several classes of legitimate event produce observations indistinguishable from attacks in isolation. A firmware update changes OS fingerprint and may change port profile and TTL baseline. A DHCP

lease renewal reassigns an IP, potentially to a different device. A service restart or version upgrade changes the service banner and may open or close ports transiently. MAC address randomisation, enabled by default in modern mobile operating systems, breaks MAC-keyed identity continuity for mobile devices. A router reconfiguration shifts TTL values and route paths simultaneously for all subnet devices. These events define the false positive ceiling: any signal that fires on them without corroboration or weighting will produce an alert rate that renders the system unusable in practice [2]. Evidence accumulation, signal weighting, and score decay are therefore requirements rather than optional refinements, they are the mechanism by which the system remains useful under legitimate network activity.

The deployment target imposes hard operational constraints. Nmap OS detection requires root privilege to send raw socket probes; promiscuous-mode packet capture requires root and a network interface that can see subnet broadcast traffic. Neither constraint can be mitigated in software. Nmap scanning a /24 subnet takes between 15 and 60 seconds depending on host density and scan depth; the minimum tick interval is bounded below by this figure. Passive capture is continuous while active scans are periodic: passive observations must be attributed to the device identity established by the most recent active scan, and devices that generate no outgoing traffic between scans are invisible to passive observation. There is no managed switching infrastructure, SNMP feed (Simple Network Management Protocol — device telemetry polling), NetFlow export (per-flow traffic metadata generated by managed routers and switches), or SPAN port (Switched Port Analyser — a switch feature that mirrors all traffic from one port to a dedicated monitoring port) available; the monitoring host sees only traffic visible on its local interface and subnet broadcasts.

Per-device threshold analysis has a structural failure mode: a coordinated attack affecting multiple devices simultaneously, each producing insufficient evidence to cross the per-device threshold individually, will produce no alert. Lowering the per-device threshold to catch such attacks would generate unacceptable false positives under legitimate fleet-wide events, a router reconfiguration shifts TTL values for all devices at once. Fleet-level correlation is therefore a first-class detection requirement rather than a sensitivity adjustment. It requires a bounded tick window within which events from multiple peers are co-evaluated; the window start is the previous tick timestamp, and only events recorded after that timestamp are eligible for fleet pattern matching.

Two architectural constraints emerge from the problem domain before any design decisions are made. First, the LLM must not receive adversarially-controlled data. Device-reported hostnames, service banners, and packet payloads are attacker- controllable strings that, if passed to the model, expose it to indirect prompt injection [13], a compromised device could craft its hostname to cause the LLM to suppress or misclassify the alert investigating it. LLM input must therefore be restricted to data derived entirely by the detection pipeline: numeric scores, event type labels drawn from a fixed enum, and device identifiers assigned by PeerWatch, not reported by the device. Second, autonomous blocking carries irreversibility risk: an incorrectly blocked device may take a legitimate service offline. The decision to block must be governed by deterministic rule-based guards that LLM output cannot override; the model's severity classification may be one guard among several, but the guard logic itself must be purely rule-based.

REFERENCES

[1] Manos Antonakakis, Tim April, Michael Bailey, Matt Bernhard, Elie Bursztein, Jaime Cochran, Zakir Durumeric, J. Alex Halderman, Luca Invernizzi, Michalis Kallitsis, and others. 2017. Understanding the Mirai Botnet. In *Proceedings of the 26th USENIX Security Symposium*, 2017. 1093–1110.

[2] Stefan Axelsson. 2000. The Base-Rate Fallacy and the Difficulty of Intrusion Detection. *ACM Transactions on Information and System Security* 3, 3 (2000), 186–205.

[3] Philippe Biondi. 2003. Scapy – Packet Manipulation Library.

[4] Franco Callegati, Walter Cerroni, and Marco Ramilli. 2009. Man-in-the-Middle Attack to the HTTPS Protocol. *IEEE Security & Privacy* 7, 1 (2009), 78–81.

[5] Harrison Chase and others. 2022. LangChain: Building Applications with Large Language Models.

[6] Cisco Systems. 2020. Cisco Annual Internet Report (2018–2023) White Paper.

[7] Cisco Systems. 2024. Cisco Secure Network Analytics.

[8] Samuel Colvin and others. 2024. Pydantic — Data Validation Using Python Type Hints.

[9] Darktrace. 2024. Darktrace Enterprise Immune System.

[10] Zakir Durumeric, David Adrian, Ariana Mirian, Michael Bailey, and J. Alex Halderman. 2015. A Search Engine Backed by Internet-Wide Scanning. In *Proceedings of the 22nd ACM SIGSAC Conference on Computer and Communications Security*, 2015. 542–553.

[11] Zakir Durumeric, Eric Wustrow, and J. Alex Halderman. 2013. ZMap: Fast Internet-Wide Scanning and Its Security Applications. In *Proceedings of the 22nd USENIX Security Symposium*, 2013. 605–620.

[12] Robert David Graham. 2013. masscan: Mass IP Port Scanner.

[13] Kai Greshake, Sahar Abdelnabi, Shailesh Mishra, Christoph Endres, Thorsten Holz, and Mario Fritz. 2023. Not What You've Signed Up For: Compromising Real-World LLM-Integrated Applications with Indirect Prompt Injection. In *Proceedings of the 16th ACM Workshop on Artificial Intelligence and Security*, 2023.

[14] Wajih Ul Hassan, Shengjian Guo, Ding Li, Zhengzhang Chen, Kangkook Jee, Zhichun Li, and Adam Bates. 2019. Nodoze: Combatting threat alert fatigue with automated provenance triage. In *network and distributed systems security symposium*, 2019.

[15] Van Jacobson, Craig Leres, and Steven McCanne. 1994. libpcap – Packet Capture Library.

[16] Craig Leres and Van Jacobson. 1992. arpwatch – Ethernet/IP Address Pairings Monitor. Retrieved from https://ee.lbl.gov/

[17] Gordon Fyodor Lyon. 2008. *Nmap Network Scanning*. Insecure.Com LLC. Retrieved from https://nmap.org/book/

[18] Keyu Man, Zhiyun Qian, Zhongjie Wang, Xiaofeng Zheng, and Youjun Huang. 2020. DNS Cache Poisoning Attack: Resurrections with Side Channels. In *Proceedings of the 2020 ACM SIGSAC Conference on Computer and Communications Security*, 2020.

[19] Markus Miettinen, Samuel Marchal, Ibbad Hafeez, N. Asokan, Ahmad-Reza Sadeghi, and Sasu Tarkoma. 2017. IoT Sentinel: Automated Device-Type Identification for Security Enforcement in IoT. In *Proceedings of the 37th IEEE International Conference on Distributed Computing Systems (ICDCS)*, 2017. 2177–2184.

[20] MITRE Corporation. 2024. MITRE ATT&CK Enterprise Tactics, Techniques and Procedures.

[21] Ollama. 2024. Ollama – Run Large Language Models Locally.

[22] Open Information Security Foundation. 2024. Suricata Network IDS, IPS and NSM Engine.

[23] KimiNewt Ouri. 2014. PyShark – Python Packet Parsing using tshark.

[24] Vern Paxson. 1998. Bro: A System for Detecting Network Intruders in Real-Time. In *Proceedings of the 7th USENIX Security Symposium*, 1998.

[25] David C. Plummer. 1982. *An Ethernet Address Resolution Protocol.* Retrieved from https://www.rfc-editor.org/rfc/rfc826

[26] Fauzan Prasetyo Eka Putra, Ubaidi Ubaidi, Alief Badrit Tamam, and Reynal Widya Efendi. 2024. Implementation And Simulation Of Dynamic Arp Inspection In Cisco Packet Tracer For Network Security. *Brilliance: Research of Artificial Intelligence* 4, 1 (2024), 340–347.

[27] Qualys Threat Research Unit. 2024. regreSSHion: Remote Unauthenticated Code Execution Vulnerability in OpenSSH Server (CVE-2024-6387).

[28] Martin Roesch. 1999. Snort: Lightweight Intrusion Detection for Networks. In *Proceedings of the 13th USENIX Conference on System Administration (LISA)*, 1999. 229–238.

[29] Dug Song and others. 2004. dpkt – Fast, Simple Packet Creation and Parsing.

[30] US-CERT. 2017. TA17-181A: Petya/NotPetya Ransomware.

[31] Wireshark Foundation. 2024. Wireshark Network Protocol Analyser.

[32] Michal Zalewski. 2012. p0f v3 – Passive OS Fingerprinter.

D ARTIFACT APPENDIX

In this section we show how to reproduce our findings.